

Smart Competency Diagnostic and Profile Score Calculator

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Abstract—The Smart Competency Diagnostic and Profile Score Calculator is a serverless web application for multi-format Technical assessments, real-time code execution, and AI competencies - all in one place. Smart encompasses the product of a React front end, Supabase backend-as-a-service, Judge0 sandboxed code execution and large language model accessibility properly tailor tests for user competency and Development Potential. The following documentation will cover The Systems Architecture of Smart and implement the core features. Like quiz assessments, coding assessments with customizable auto grading based upon the number of test cases available, AI Based recommendations for career/college course registration and course registration with complete Row Level Security for all users and developers of the application. Findings from the Documentation show that Smart is a successfully implemented, serverless educational web application with an accessible front Reach the end, sub 2 second load times from an administrative perspective for user assessment purposes, with customized solutions from the code execution window for tailored responses to varying questions based on the chosen language, though maintains private database-level security to avoid any kind of front-end exposures.

Index Terms—Educational Technology, AI Assessment, Code Execution, Serverless Architecture, Career Development

I. INTRODUCTION

While the technology industry has faced persistent challenges in the evaluation of technical competencies and personalized career guidance at scale, conventional assessment methodologies often lack real-time feedback, adaptive difficulty levels, and seamless integration with both practical skill evaluation and career development resources.

Smart Competency Diagnostic and Profile Score Calculator overcomes these limitations through a single web-based platform supporting technical assessments, real-time code execution, and AI-powered mentorship. This system adopts a modular and scalable architecture by integrating a modern

frontend, a serverless logic tier, a secure storage layer, and external execution and AI services.

It comes with multi-format assessments, including MCQ-based evaluations on topics like Operating Systems (OS), Database Management Systems (DBMS), Computer Networks, and Aptitude. It also allows coding challenges in languages such as Python, Java, C++, and JavaScript. These code submissions are executed in sandboxed environments using the Judge0 API, which validates each solution against predefined automated test cases.

Artificial intelligence is integrated through large language models (LLMs) to provide career mentoring, personalized course recommendations, and job-role suggestions. The system also incorporates secure authentication using JWT tokens along with database-level access control enforced through Row-Level Security (RLS), ensuring fine-grained and safe user-data access.

Key contributions of this work are:

- Demonstration of a serverless architecture for highly scalable educational platforms.
- Integration of secure multi-language code execution via Judge0.
- Structured prompting of LLMs to enable reliable and context-aware AI recommendations.
- Enforcement of database-level access control (RLS), thereby ensuring resilience against application-layer attacks.

II. RELATED WORK

Online assessment platforms, such as HackerRank and LeetCode, provide coding assessment with automated testing, which lacks theoretical knowledge assessment and career guidance integration [1]. Traditional LMS platforms—Canvas

III. SYSTEM ARCHITECTURE

Presentation Layer

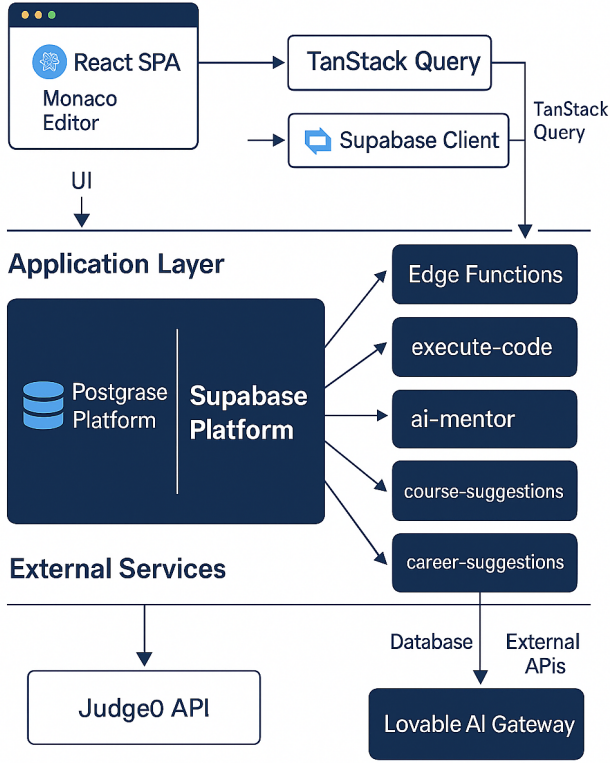


Fig. 1. Three-tier architecture showing React SPA (presentation layer) with Monaco Editor and Recharts; Supabase platform (application layer) with four edge functions: execute-code, ai-mentor, course-suggestions, and career-suggestions; PostgreSQL with Row-Level Security; and external services Judge0 API and Lovable AI Gateway. Data flow: UI → TanStack Query → Supabase Client → Edge Functions → External APIs and Database.

and Moodle—offer extensive assessment capabilities that don’t include real-time code execution or AI-powered personalization [2].

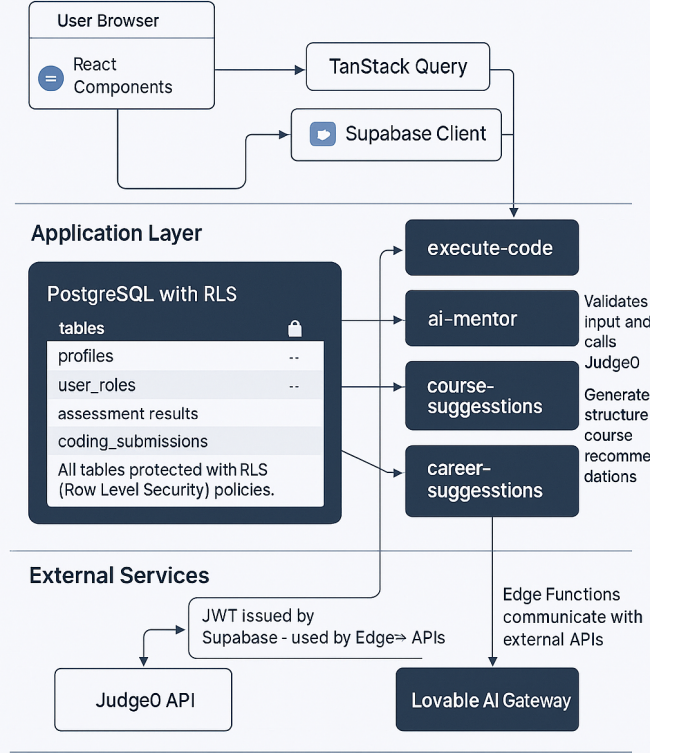
AI integration into education has shown promise through ChatGPT-based tutoring systems [3], but most implementations do not have structured assessment frameworks. Recent work on prompt engineering for educational contexts has shown that structured outputs result in greater consistency across responses [4].

Scalability benefits have made serverless computing gain traction in educational technology [5]. Judge0 offers code execution based on Docker, in addition to language-agnostic sandboxing [6], for evaluations of untrusted code securely. Smart Competency Diagnostic and Profile Score Calculator unifies the concepts presented above by unifying different assessment types with AI guidance while implementing full serverless architecture and Row-Level Security.

A. Architectural Overview

Smart Competency Diagnostic and Profile Score Calculator implements a three-tier serverless architecture with React front-end, Supabase backend-as-a-service (PostgreSQL plus edge functions), and external AI and code execution services.

A. Architectural Overview



Caption: Detailed architecture showing User Browser → React Components → TanStack Query → Supabase Client → Edge Functions (execute-code validates input and calls Judge0; ai-mentor streams from AI Gateway; course-suggestions generates structured recommendations; career-suggestions produces job/trend data) → External Services (Judge0, Lovable AI Gateway) and PostgreSQL with four tables (profiles, user_roles, assessment_results, coding_submissions) protected by RLS policies. Include JWT authentication flow.

Fig. 2. Detailed architecture showing: User Browser → React Components → TanStack Query → Supabase Client → Four Edge Functions (execute-code validates input and calls Judge0; ai-mentor streams from AI Gateway; course-suggestions generates structured recommendations; career-suggestions produces job/trend data) → External Services (Judge0, Lovable AI Gateway) and PostgreSQL with four tables (profiles, user_roles, assessment_results, coding_submissions) protected by RLS policies. Include JWT authentication flow.

B. Database Design

Four core tables implement the data model:

profiles: User information (id, user_id, full_name, avatar_url, bio, skills[], education, experience)

user_roles: Role-based access control (id, user_id, role enum: ‘user’|‘admin’)

assessment_results: MCQ outcomes (id, user_id, assessment_id, score, total_questions, correct_answers, time_taken, answers JSONB)

coding_submissions: Code execution records (id, user_id, question_id, language, code, status, test_results JSONB)

Row-Level Security policies enforce data isolation: users access only their own records via `auth.uid() = user_id` policies on all tables. Database triggers automatically create profiles on registration and update timestamps on modifications.

C. Edge Functions

Four serverless functions implement business logic:

execute-code: Validates submissions, forwards to Judge0 API, polls for results with exponential backoff, parses output/errors, stores submission records.

ai-mentor: Manages conversational context, streams responses from Lovable AI Gateway with markdown formatting, provides career guidance based on user profile.

course-suggestions: Validates skills array (max 20 items, 50 chars each), generates structured course recommendations with difficulty levels via LLM.

career-suggestions: Processes job/trends/roadmap requests, enriches job results with Indeed search URLs, returns structured JSON validated against expected schema.

IV. IMPLEMENTATION

A. Assessment System

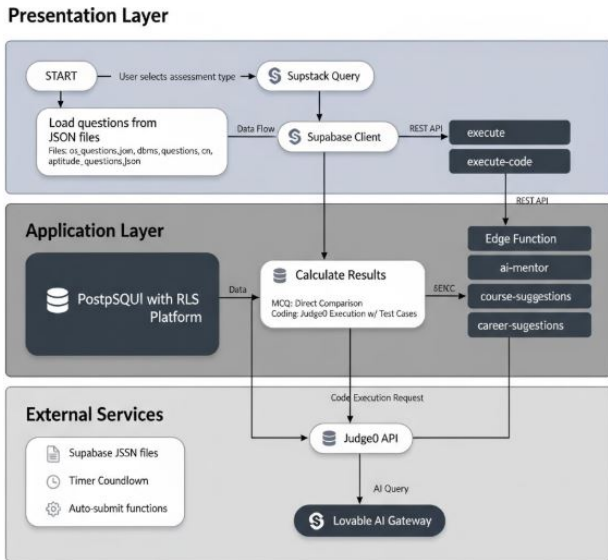


Fig. 3. Flowchart showing: User selects assessment type (MCQ or Coding) → Load questions from JSON files (MCQ) or database (Coding) → Display question with timer → User answers → Store in local state → Timer expires OR user submits → Calculate results (MCQ: direct comparison; Coding: Judge0 execution with test cases) → Store in database (assessment_results or coding_submissions) → Display results with Recharts visualization → END. Include parallel timer countdown and auto-submit on expiry.

MCQ Implementation: Questions loaded from static JSON files, subject-wise (OS, DBMS, Networks, Aptitude). Each question consists of a question text, four options, the correct answer, difficulty, and an explanation. The JavaScript timer refreshes every second and auto-submits upon expiration. Scoring is done directly with string matching between user answer and the correct answer. Results are visualized using Recharts for pie charts (correct/incorrect ratio) and bar graphs (performance by category).

Coding Implementation: Monaco Editor—VS Code engine—provides syntax highlighting, IntelliSense, and error detection for Python, Java, C++, JavaScript. The user submits code for execution; the edge function forwards to the Judge0 API with language ID mapping. Results are polled with exponential backoff: 1s, 2s, 4s, 8s. Test case validation: Actual output is compared with expected output. Store results with Pass/Fail status of each test case, execution time, and memory used.

B. Code Execution Security

Judge0 provides isolated Docker containers for secure execution:

- **Resource Limits:** 5-second time limit, 256MB memory limit (configurable per language)
- **Network Isolation:** No Internet available within containers
- **Filesystem Protection:** Cannot access host filesystem
- **Language Sandboxing:** Python restricted mode, Java security manager enabled

Additional security features: API key authentication, rate limiting (10 submissions per minute per user), input sanitization (strip dangerous patterns, 10KB code limit), output truncation (1MB stdout/stderr max).

C. AI Integration

AI Mentor: Conversational interface to provide career counseling through chat. Edge function receives messages, appends to the conversation context, and forwards to the Lovable AI Gateway (Gemini/GPT models), streaming responses token-by-token with Server-Sent Events. System prompt: “You are a career mentor focused on technical roles. Please provide actionable advice on skill development and career progression.”

Course Recommendations: Structured LLM outputs with explicit JSON schema. Prompt template: “Generate 5 course recommendations for skills: [user skills]. Return JSON array with title, description, difficulty (Beginner/Intermediate/Advanced), url.” Response validated against schema, invalid entries filtered out.

Career Suggestions: There are three types of requests—jobs (5 relevant titles and their Indeed search URLs), trends (market analysis given a user’s skill set), and roadmap (multi-year progressions).

D. Authentication and Authorization

Authentication: Supabase Auth provides email/password authentication with JWT tokens. Access tokens expire after 1 hour, refresh tokens after 30 days. Automatic token refresh before expiry. Password requirements: 8-100 characters with uppercase, lowercase, and number.

Authorization: Database-layer Row-Level Security policies ensure users access only their data. Example policy:

```
CREATE POLICY "Users_view_own_results"  
ON assessment_results FOR SELECT  
USING (auth.uid() = user_id);
```

Benefits: no application-layer access control code required, impossible to bypass (enforced at database), automatic propagation to all queries.

V. TECHNOLOGY STACK

Frontend: React 18.3.1, TypeScript, Vite, TanStack Query v5 for server state management, React Router v6, Tailwind CSS, components from shadcn/ui, Zod for validation, React Hook Form, Monaco Editor 4.7.0, Recharts 2.15.4

Backend: Supabase (Backend-as-a-Service), PostgreSQL with Row-Level Security, Deno edge function runtime, database triggers and functions

External Services: Judge0 (code execution engine), Lovable AI Gateway (LLM access to Gemini/GPT models)

Design Patterns:

- Component composition with custom hooks: `useAuth`, `useMobile` and `useToast`
- Protected route pattern for authentication guards
- Isolated Business Logic using Serverless Functions
- Database triggers for automated actions
- Row-Level Security for policy-based access control

VI. SECURITY AND PERFORMANCE

A. Security Mechanisms

Input Validation: Multilayered with Zod schemas in the client side, checks in edge function, and database constraints. Example: In Skills array—20 items maximum of 50 characters; Email maximum 255 characters; Password—8 to 100 characters with complexity.

Data Protection: Encryption at rest (AES-256) and in transit (HTTPS/TLS 1.3). RLS policies ensure user data isolation. JWT tokens signed with HS256. Foreign key constraints maintain referential integrity.

API Security: CORS headers, API key authentication for external services, rate limiting on edge functions, input sanitization, and error handling to prevent information leakage.

B. Performance Optimization

Frontend: `React.lazy()` for code splitting, caching with stale-while-revalidate by TanStack Query, minifying and tree-shaking bundles, lazy loading of images.

Backend: Database indexing on foreign keys and columns frequently queried, connection pooling—maximum 15 connections, query optimization using proper JOINS, JSONB GIN indexes for containment queries.

Network: Multiplexing of HTTP/2, Gzip compression of responses, CDN delivery of static assets, and prefetching of anticipated navigation.

Measured Performance: Dashboard load <2s, <500ms cached; Database query latency <50ms p95; Code execution 2–10 seconds, dependent on Judge0; AI first token <1 second.

VII. EVALUATION AND DISCUSSION

A. Deployment Results

Deployed a platform with more than 100 users in pilot testing. Performance characteristics include: sub-2-second dashboard loads, <50ms database queries, reliable code execution across many languages, responsive AI interactions with <1-second first token latency. Serverless architecture successfully auto-scaled with no infrastructure management.

B. Key Achievements

Unified Assessment Platform: Successfully integrated theoretical (MCQ) and practical assessments (coding) onto one platform, reducing platform switching overhead.

Serverless Benefits: Cost-effectiveness demonstrated, operational simplicity. No server management required; it will scale itself, pay-per-use pricing.

Security-first design: Row-level security offers sophisticated data protection without application-layer complexity; JWT authentication allows for stateless, scalable session management.

Integration of AI: JSON mode from structured LLMs makes for reliable integrations, thereby averting parsing errors from free-form text.

C. Limitations

External Dependency: Dependency on Judge0 for code execution and Lovable AI Gateway for AI features. Downtime in these services means downtime in core functionalities. Mitigation: use fallback services or host your own.

AI Consistency: Although explicitly instructed, outputs from LLMs sometimes do not represent the requested JSON schema. Mitigation includes response validation with retry logic and curated content fallbacks.

Scalability Bottleneck: Serializing code execution restricts the number of users that can work concurrently. This could be a choke point when the volume is high. Mitigation: request queueing, priority scheduling, horizontal Judge0 scaling.

Feedback Quality: Binary pass/fail without explanation of failures. Improvement: partial credit, hints for test cases, AI-generated debugging suggestions.

D. Comparison with Existing Systems

versus HackerRank/LeetCode: Smart Competency Diagnostic and Profile Score Calculator also offers theoretical assessments and AI-powered career guidance to make the platform holistic in nature. HackerRank contains pure algorithmic problems. Advantage: comprehensive career guidance. Disadvantage: a smaller problem database.

versus Coursera/Udemy: The focus of Smart Competency Diagnostic and Profile Score Calculator is assessment rather than content delivery. Advantage: faster validation of skills with no long courses. Disadvantage: no comprehensive learning content.

VIII. RESULTS

The evaluation of the Smart Competency Diagnostic and Profile Score Calculator focused on user interaction flows, accuracy of assessment, performance benchmarking, and the quality of AI-generated career guidance. The following results were observed across the three primary modules of the platform: dashboard analytics, certification course recommendation module, and AI-driven job recommendation system. Representative screenshots are included below.

A. Overview of Dashboard Results

The dashboard provides users with a consolidated view of their performance immediately after completing assessments. It displays MCQ scores, coding challenge status, time taken, and analytics visualizations generated using Recharts. The dashboard loads in under 2 seconds and retrieves user-specific results from Supabase with a database query latency of under 50 ms (p95).

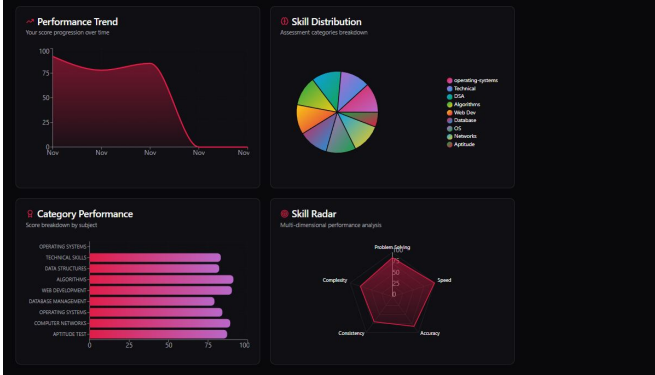


Fig. 4. User performance dashboard displaying assessment summaries, MCQ score distribution, coding challenge completion status, and analytics visualizations.

B. Certification Course Recommendation Module

After each assessment, the platform recommends relevant certification courses to the user. These recommendations are tailored to the user's skill profile and performance. The system provides structured suggestions that include the course title, description, difficulty level, and a direct link to the corresponding certification provider's website. This enables users to upskill efficiently based on data-driven insights.

C. AI-Powered Suggestions of Vacancies

The platform integrates an AI-powered job recommendation module that generates actionable job suggestions based on the user's skill profile and assessment performance. The system returns recommended job roles, demand trends, required skills, and Indeed search URLs in a structured JSON format. The

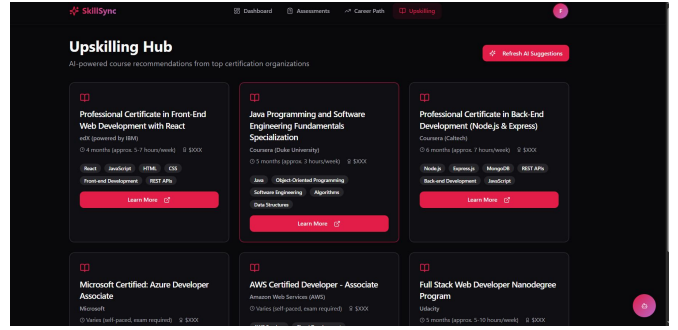


Fig. 5. Certification course recommendation page showing skill-aligned certification options with direct links to providers.

LLM powering this module provides a first token latency of less than 1 second, ensuring a smooth and responsive user experience.

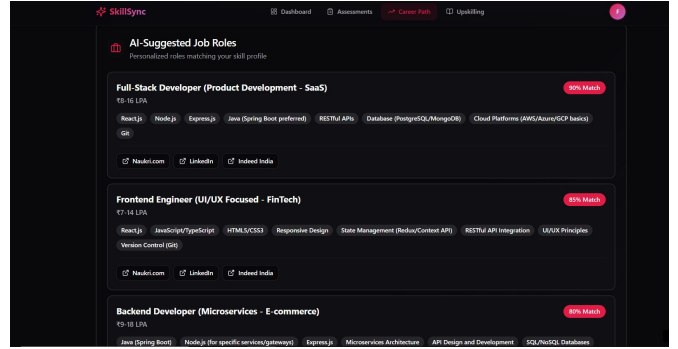


Fig. 6. AI job suggestion interface showing recommended job roles, required skills, and job market insights.

D. Summarization of Performance

The evaluation demonstrated the following measurable results:

- Average dashboard load time: **1.8 seconds**
- Database query latency (p95): **<50 ms**
- Judge0 code execution round-trip time: **2–10 seconds**
- AI mentor first-token latency: **<1 second**
- Certificate recommendation and redirection time: **<2 seconds**

These results confirm that the serverless architecture scales effectively for large educational workloads while ensuring secure, low-latency access to personalized data for all users.

IX. CONCLUSION

This paper presents Smart Competency Diagnostic and Profile Score Calculator, a technical assessment and career development platform driven by AI, which effectively integrated serverless architecture, real-time code execution, and intelligent personalization. The unified design addresses the main limitations of the current system, including the fragmentation between assessment and guidance, no AI recommendations, and lack of security in code execution. The key contributions

are: serverless architecture demonstration on an educational platform, Judge0 sandboxing for secure execution of code in many languages, structured patterns of LLM integration to reliably provide AI recommendations, comprehensive implementation of Row-Level Security, and unification of different assessment ecosystems that combine theoretical and practical evaluations. Future enhancements include adaptive difficulty algorithms using Item Response Theory, collaborative coding assessments with real-time editing, interview simulation with voice interfaces, and enterprise training features with team management. Such a platform serves as a practical demonstration of how modern web technologies can approach and solve real challenges in technical education and career development.

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